

Driver: Online and Offline Modes

Problem Alignment: DispatchOne currently does not record operational hours of a driver, which is required for compliance purposes. Hence, this document scopes the solution for capturing offline and online hours of a driver which can be used for compliance requirements and work as a means for driver's attendance.

System Design:

1. New substates under "Active" will be introduced, and their state transitions are as shown below in Tables 1.1 and 1.2.
2. Ability to Go Online and Go Offline will be introduced in the Driver Mobile App via a Mark IN/Out button.
3. A flag will be introduced at the tenant level to decide if drivers of that tenant will have access to this functionality.
4. The default state of the driver after onboarding will be active:available-offline for tenants when the flag is enabled.
5. The default state of the driver after onboarding will be active:available-online for tenants when the flag is disabled.
6. Online/offline time is stored for every driver in the data warehouse.
7. A default shift duration is set up for every tenant in the backend (Example: 9 hours for Bira). This shift duration will be used for initiating the fallback logic of "Auto Punch Out" and will be replaced with the roster of a driver (when Roster FPA is designed).
8. Dispatch cannot be started when the driver assigned is offline.
9. Green dot will be used to represent online mode and orange dot will represent offline mode on UI.

Online/Offline flow on the Driver App: [Figma Screens Link](#)

Online/Offline flow on the Web App: [Figma Screens Link](#)

State Transition: State changes of the driver based on the online/ offline modes are:

Table 1.1 – Flag is enabled

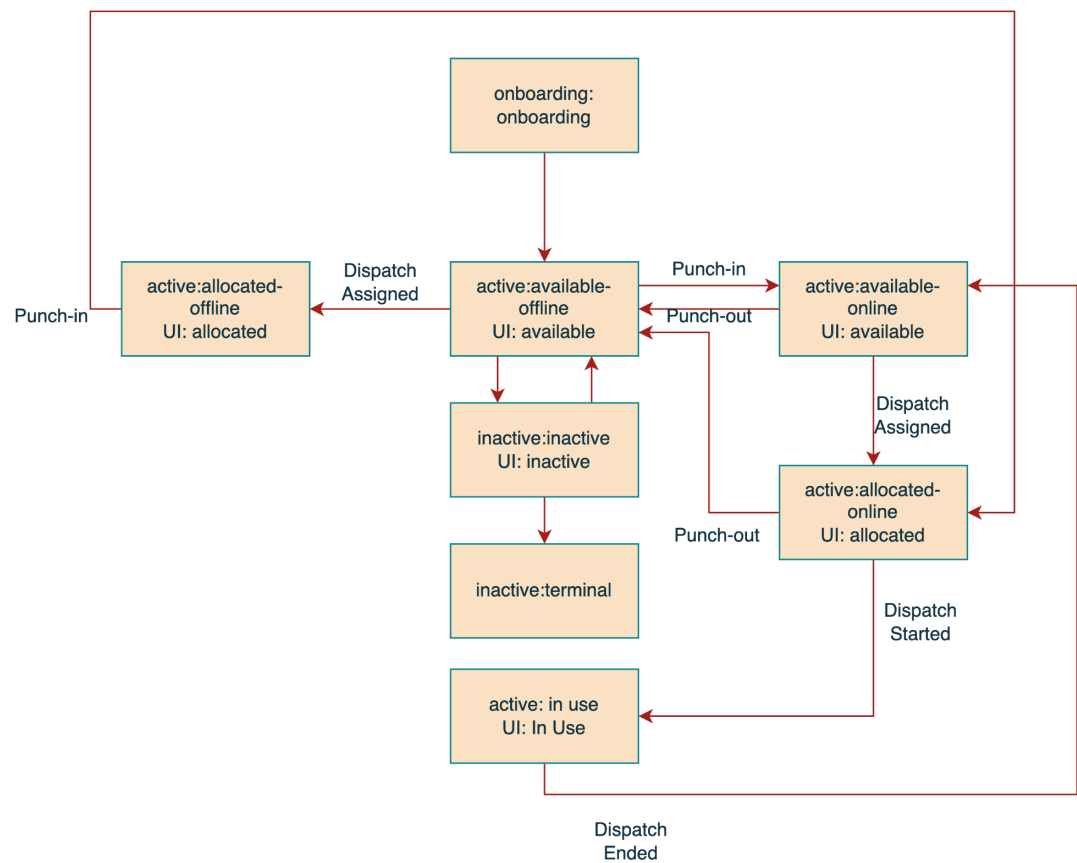
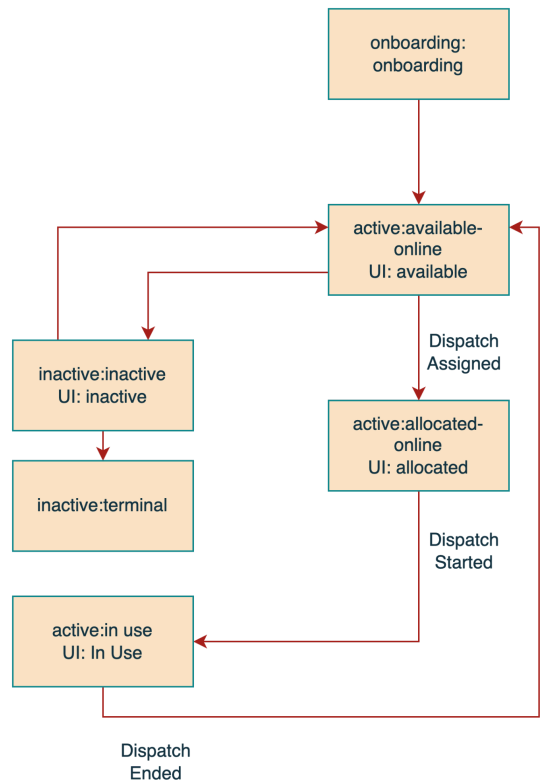


Table 1.2 – Flag is disabled



Proposed Solution:

1. Driver needs to Punch In after login by clicking on the "Go Online" button in the Driver App if not already Punched In. The "Go Online" button is shown on the home page. OSI captures the time, date and location (lat/long) of the punch.
2. Driver needs to Punch Out by clicking on the "Go Offline" button after the end of a dispatch. The "Go Offline" button is shown on the homepage when a driver has Punched In. OSI captures the time, date and location (lat/long) of the punch.
3. If not Punched Out by the end of shift duration, OSI will automatically initiate a Punch Out and record the time as follows.
 - A. If there is an ongoing dispatch, then the offline activation time would be the end of dispatch timestamp.
 - B. If there is no ongoing dispatch, then the offline activation time would be online activation time + shift duration.

Scenario 1: Driver without an associated vehicle.

1. Driver FPA:
 - A. Online/offline status is reflected next to the driver's name on the driver list page and driver details page.
 - B. Add a new filter for "Mode" with two labels (Online, Offline) on all, available, and allocated tabs.
 - C. Colour of status label: The colour of the allocated status label is to be teal and of the in use status label to be orange.
2. Order Pages:
 - A. List View: For the 'Create Manual Dispatch' and 'Create Automatic Dispatch' modals, the driver status will be highlighted in the driver dropdown.
 - B. Map View: For the 'Plan by vehicle' tab of 'Create Automatic Dispatch' and 'Create Manual Dispatch' the driver status will be highlighted in the driver dropdown. The driver status will also be highlighted under the dispatches tab.
3. Dispatch Pages:
 - A. Driver status will be highlighted in the driver dropdown shown in the 'Create Dispatch' modal along with 'Edit Vehicle/Rider' modal under the assigned, ongoing, and pending closure states of Dispatch List page and in the dispatch details page.
 - B. When the driver assigned to a dispatch is offline, then the start dispatch button is disabled.

4. Route Planner Page: In the routes tab, under the 'Edit Vehicle/driver' modal, driver status will be highlighted in the driver dropdown.
 - A. When the driver assigned to a dispatch is offline, then the assign & start dispatch button is disabled.
5. Real Time Dashboard: Add two text tabs for the count of drivers in the online and offline modes under driver widget. The count of available, allocated and dispatched based on the driver states is shared below.
 - A. Available (available:online + available:offline)
 - B. Allocated (allocated:online + allocated:offline)
 - C. Dispatched (in use)
6. Analytics FPA: The calculation of the "Total active drivers" field would be updated. The new calculation would be the sum of drivers in these states (available:offline + available:online + allocated:offline + allocated:online + in use) in a day.

Scenario 2: Driver associated with a vehicle. This association is established when a vehicle ownership is tagged with an operator (Driver).

1. Vehicle FPA Page:
 - A. Text edits: Existing "AVAILABLE" status to have a sentence case "Available". Terminated status to be renamed as "Inactive" and Inuse status to be renamed as "Allocated".
 - B. Colour of status label: The colour of the allocated status label should be teal.
2. Order Pages:
 - A. List View: For the 'Create Manual Dispatch' and 'Create Automatic Dispatch' modals, the driver status will be highlighted in the dropdowns.
 - i. The vehicle's number is shown next to the driver's name and the driver's name is shown next to the vehicle's number in the dropdowns.
 - ii. When a driver is selected from the driver dropdown, then the associated vehicle number is recommended and is shown as the first option in the vehicle dropdown.
 - iii. When a vehicle is selected from the vehicle dropdown, then the associated driver name is recommended and is shown as the first option in the vehicle dropdown.

B. Map View: For the 'Plan by vehicle' tab of 'Create Automatic Dispatch', 'Create Manual Dispatch' and 'Create Automatic Dispatch' modals, the driver status will be highlighted in the dropdowns.

- i. The vehicle's number is shown next to the driver's name and the driver's name is shown next to the vehicle's number in the dropdowns.
- ii. When a driver is selected from the driver dropdown, then the associated vehicle number is recommended and is shown as the first option in the vehicle dropdown.
- iii. When a vehicle is selected from the vehicle dropdown, then the associated driver name is recommended and is shown as the first option in the vehicle dropdown.
- iv. The driver and vehicle statuses will be highlighted under the dispatches tab.

3. Dispatch Pages:

A. Dispatch List Page: Driver status will be highlighted in the dropdowns shown in the 'Create Dispatch' modal along with 'Edit Vehicle/Rider' modal under the assigned, ongoing, and pending closure states of Dispatch List page. Driver status will also be highlighted on the dispatch details page.

- i. The vehicle's number is shown next to the driver's name and the driver's name is shown next to the vehicle's number in the dropdowns.
- ii. When a driver is selected from the driver dropdown, then the associated vehicle number is recommended and is shown as the first option in the vehicle dropdown.
- iii. When a vehicle is selected from the vehicle dropdown, then the associated driver name is recommended and is shown as the first option in the vehicle dropdown.

4. Route Planner Page: In the routes tab, under the 'Edit Vehicle/driver' modal, driver status will be highlighted in the dropdowns.

A. The vehicle's number is shown next to the driver's name and the driver's name is shown next to the vehicle's number in the dropdowns.

ADO Link:

https://dev.azure.com/DelhiveryPlatform/OSI-Applications/_workitems/edit/22258